Executive Summary

Customer Segmentation Analysis

Prepared for: Marketing & Growth Leadership
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Objective

Identify distinct customer groups within the retail customer base so marketing spend can be targeted rather than blanket, improving retention and revenue efficiency.

Approach

We analyzed 1,200 customers (built on a real Kaggle customer dataset, expanded with modeled behavioral data) across seven behavioral dimensions: age, income, spending score, purchase frequency, recency, tenure, and monetary value. Three clustering algorithms — K-Means, Hierarchical Clustering, and DBSCAN — were tested and compared. K-Means (k=4) was selected as the production model based on cluster balance, interpretability, and stability testing.

The Four Segments

Segment	Share	Profile	Recommended Action
Premium Spenders	21%	Highest income & spending, most frequent, most recent, longest tenure	VIP loyalty tier, priority support, referral program
Steady Mature Regulars	24%	Older, mid income/spend, dependable	Cross-sell campaigns, mid-tier loyalty perks
New Bargain Hunters	28%	Younger, price-sensitive, engaged, growing	Time-boxed discounts, bundle deals, rewards
At-Risk Customers	27%	High income but lowest engagement, ~120 days since last order	Urgent win-back campaign, root-cause survey

Key Findings

• **No single metric predicts value.** Income alone does not predict spending — multi-dimensional clustering was necessary to find real structure.

• **At-Risk Customers are the highest-leverage opportunity.** They have the income to spend but have quietly disengaged — unlike low-income segments, the barrier isn't affordability, making them a strong win-back target.

• **The segmentation is statistically stable.** Repeated testing on random subsamples showed consistent results, meaning these segments can be trusted for ongoing campaign planning, not just a one-time snapshot.

• **DBSCAN was evaluated but rejected for production use** — while it scored well numerically, it produced one oversized cluster and many tiny, unusable fragments rather than clean, equal-sized business segments.

Recommended Next Steps

1. Launch a win-back campaign for At-Risk Customers within the next marketing cycle.
2. Pilot a VIP loyalty tier for Premium Spenders to protect and grow this high-value segment.
3. Re-run this segmentation quarterly and track segment migration over time.
4. Replace engineered behavioral fields with real transaction data as it becomes available.

Expected Business Impact

Targeted campaigns per segment (instead of one blanket strategy) are expected to improve campaign response rates and reduce wasted spend on customers unlikely to respond to generic offers, while directly addressing the revenue at risk in the At-Risk segment.

Full technical methodology, code, and visualizations are available in the accompanying Jupyter notebook and GitHub repository.